# TYPES OF DATA ANALYSIS

30-Page Academic Module

Data Science / Data Analytics

A structured learning module covering the major approaches used to understand, explain, predict, and support decisions from data.

# 1. Introduction to Data Analysis

## Overview

Data analysis is the systematic process of collecting, cleaning, transforming, exploring, interpreting, and communicating data to answer questions and support decisions. It converts raw observations into useful information and, ultimately, actionable knowledge.

## Why Data Analysis Matters

Organizations generate data from transactions, sensors, surveys, websites, applications, experiments, and social platforms. Analysis helps identify patterns, measure performance, understand customers, evaluate hypotheses, detect problems, and predict future outcomes.

## Basic Analysis Cycle

A typical workflow is: define the problem → collect data → clean data → explore data → apply analytical methods → interpret results → communicate findings → take action.

## Example

A university can analyze attendance, assessment scores, and course feedback to identify subjects where students need additional support.

# 2. Major Types of Data Analysis

## Main Categories

Data analysis is commonly divided into descriptive, diagnostic, exploratory, inferential, predictive, and prescriptive analysis. Different classifications also include causal, qualitative, quantitative, and real-time analysis.

| Type | Main Question | Typical Output |
|---|---|---|
| Descriptive | What happened? | Reports, summaries, dashboards |
| Diagnostic | Why did it happen? | Drivers, relationships, root causes |
| Exploratory | What patterns exist? | Trends, clusters, anomalies |
| Inferential | What can we conclude about a population? | Estimates, confidence intervals, tests |
| Predictive | What is likely to happen? | Forecasts, probabilities |
| Prescriptive | What should we do? | Recommendations, optimized actions |

## Important Point

These types are not mutually exclusive. A real project may use several methods in sequence.

# 3. Descriptive Data Analysis

## Definition

Descriptive analysis summarizes historical or current data so that its important characteristics can be understood quickly. It does not primarily attempt to explain causation or predict the future.

## Common Techniques

Counts, percentages, totals, averages, medians, minimum and maximum values, frequency distributions, cross-tabulations, and time-based summaries are common techniques.

## Example

For a class of 100 students, descriptive analysis can report average marks, pass percentage, highest score, lowest score, and attendance distribution.

## Advantages

It is easy to understand, useful for reporting, and provides the foundation for more advanced analysis.

# 4. Diagnostic Data Analysis

## Definition

Diagnostic analysis investigates reasons behind observed outcomes. It moves beyond 'what happened?' to ask 'why did it happen?'

## Methods

Drill-down analysis, comparison of segments, correlation analysis, root-cause analysis, variance analysis, and examination of contributing factors are commonly used.

## Example

If sales decreased in one quarter, analysts may compare regions, products, prices, promotions, inventory levels, and customer segments to identify likely causes.

## Caution

Association does not automatically establish causation. A relationship between two variables may be caused by another factor.

# 5. Exploratory Data Analysis (EDA)

## Definition

EDA is an open-ended investigation of data intended to discover patterns, relationships, unusual observations, and useful questions before formal modeling.

## Typical Activities

Inspecting distributions, plotting variables, checking missing values, examining correlations, identifying outliers, and comparing groups are central EDA activities.

## Why It Is Important

EDA can reveal data-quality problems and unexpected patterns that would otherwise produce misleading models.

## Example

A histogram of delivery times may reveal a long right tail, while a box plot may identify unusually delayed orders.

# 6. Inferential Data Analysis

## Definition

Inferential analysis uses a sample to draw conclusions about a larger population while accounting for uncertainty.

## Key Concepts

Sampling distributions, confidence intervals, hypothesis testing, p-values, standard errors, and statistical significance are important concepts.

## Example

A university surveys 300 students and uses the sample to estimate the proportion of all students who prefer online learning.

## Limitation

Inference depends on appropriate sampling, assumptions, measurement quality, and correct statistical procedures.

# 7. Predictive Data Analysis

## Definition

Predictive analysis uses historical data, statistical methods, or machine learning to estimate future or unknown outcomes.

## Applications

Demand forecasting, credit-risk scoring, customer churn prediction, fraud detection, equipment failure prediction, and recommendation systems are common applications.

## Methods

Regression, classification, time-series forecasting, decision trees, random forests, gradient boosting, and neural networks can be used.

## Example

A telecom company may predict which customers are likely to cancel their subscriptions next month.

# 8. Prescriptive Data Analysis

## Definition

Prescriptive analysis recommends actions by combining predictions, constraints, objectives, and possible decisions.

## Typical Methods

Optimization, simulation, decision analysis, scenario analysis, and rule-based recommendation systems are common approaches.

## Example

After predicting demand for several products, a retailer can optimize inventory quantities subject to storage and budget constraints.

## Difference

Predictive analysis estimates what may happen; prescriptive analysis considers what action should be taken.

# 9. Qualitative Data Analysis

## Definition

Qualitative analysis examines non-numerical information such as interview transcripts, observations, open-ended survey responses, documents, and focus-group discussions.

## Techniques

Coding, categorization, thematic analysis, content analysis, narrative analysis, and discourse analysis are widely used.

## Example

Students' written feedback can be coded into themes such as teaching quality, workload, assessment difficulty, and laboratory facilities.

## Strength

It provides context, explanations, experiences, and meanings that may not be captured by numerical variables.

# 10. Quantitative Data Analysis

## Definition

Quantitative analysis works with numerical data and applies mathematical and statistical techniques to measure relationships, differences, trends, and uncertainty.

## Examples of Data

Age, income, temperature, test scores, sales, distance, response time, and number of transactions are quantitative variables.

## Methods

Descriptive statistics, correlation, regression, hypothesis testing, ANOVA, forecasting, and machine learning are examples.

## Example

An analyst can test whether study hours are associated with examination marks using correlation or regression.

# 11. Univariate Analysis

## Definition

Univariate analysis studies one variable at a time. Its objective is to understand the variable's distribution, center, spread, and unusual values.

## Numerical Measures

Mean, median, mode, range, variance, standard deviation, quartiles, and percentiles are commonly used.

## Visualizations

Histograms, box plots, bar charts, and frequency tables are useful depending on the variable type.

## Example

Analyzing only students' examination scores can reveal the typical score and how widely scores vary.

# 12. Bivariate Analysis

## Definition

Bivariate analysis examines two variables together to understand whether and how they are related.

## Common Methods

Correlation, cross-tabulation, comparison of means, simple regression, and two-variable visualizations are common.

## Examples

Study hours versus marks, advertising expenditure versus sales, and age group versus product preference are bivariate questions.

## Important Note

The choice of method depends on whether variables are numerical, categorical, ordinal, or a mixture.

# 13. Multivariate Analysis

## Definition

Multivariate analysis examines multiple variables simultaneously. It is useful when outcomes are influenced by several factors.

## Methods

Multiple regression, logistic regression, MANOVA, factor analysis, principal component analysis, clustering, and many machine-learning methods can be used.

## Example

Predicting student performance may involve attendance, study hours, previous marks, assignment scores, and participation.

## Benefit

Considering several variables can provide a more realistic representation of complex systems.

# 14. Statistical Analysis

## Definition

Statistical analysis applies mathematical and probability-based techniques to summarize data, estimate unknown quantities, test hypotheses, and quantify uncertainty.

## Two Broad Areas

Descriptive statistics summarizes observed data, while inferential statistics uses samples to make conclusions about populations.

## Key Measures

Mean, variance, standard deviation, correlation, confidence intervals, effect sizes, and test statistics are common.

## Good Practice

Statistical conclusions should include assumptions, uncertainty, effect size, and practical relevance rather than relying only on a p-value.

# 15. Time-Series Analysis

## Definition

Time-series analysis studies observations ordered in time, such as daily sales, monthly rainfall, hourly temperature, or yearly enrollment.

## Components

Trend represents long-term movement; seasonality represents repeating patterns; cycles represent longer fluctuations; noise represents irregular variation.

## Methods

Moving averages, exponential smoothing, ARIMA-family models, decomposition, and machine-learning forecasting are widely used.

## Example

A store can forecast next month's sales using historical monthly sales and seasonal patterns.

# 16. Correlation Analysis

## Definition

Correlation measures the degree and direction of association between variables.

## Interpretation

A positive correlation means variables tend to increase together; a negative correlation means one tends to increase as the other decreases. A value near zero indicates weak linear association.

## Important Warning

Correlation does not prove causation. Confounding variables and nonlinear relationships can make interpretation difficult.

## Example

Temperature and ice-cream sales may be positively correlated, but temperature is not the only factor affecting sales.

# 17. Regression Analysis

## Definition

Regression models the relationship between an outcome variable and one or more explanatory variables.

## Types

Simple linear regression uses one predictor; multiple linear regression uses several predictors; logistic regression is commonly used for binary outcomes.

## Applications

Forecasting sales, estimating costs, understanding drivers of performance, and predicting probabilities are common uses.

## Example

A company can estimate sales from advertising expenditure, price, season, and number of active stores.

# 18. Classification Analysis

## Definition

Classification assigns observations to predefined categories or classes.

## Examples

Spam versus non-spam, pass versus fail, fraudulent versus legitimate, and high-risk versus low-risk are classification problems.

## Methods

Decision trees, logistic regression, random forests, support vector machines, gradient boosting, and neural networks are commonly used.

## Evaluation

Accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices help evaluate classification models.

# 19. Cluster Analysis

## Definition

Clustering groups observations so that items within the same group are relatively similar and items in different groups are relatively different.

## Common Methods

K-means, hierarchical clustering, DBSCAN, and model-based clustering are examples.

## Applications

Customer segmentation, document grouping, image analysis, and anomaly exploration can use clustering.

## Example

A retailer can group customers based on purchase frequency, average order value, and product preferences.

# 20. Factor and Dimensionality Analysis

## Definition

Dimensionality-reduction techniques transform many variables into a smaller set of informative components while attempting to preserve important structure.

## PCA

Principal Component Analysis creates new orthogonal components ordered by the amount of variance they explain.

## Factor Analysis

Factor analysis attempts to identify latent factors that may explain correlations among observed variables.

## Use

These methods can simplify high-dimensional datasets, reduce redundancy, support visualization, and sometimes improve modeling efficiency.

# 21. Causal Data Analysis

## Definition

Causal analysis seeks to determine whether changing one factor produces a change in an outcome.

## Approaches

Randomized experiments are a strong approach. Observational causal analysis may use matching, regression adjustment, instrumental variables, difference-in-differences, or causal graphs.

## Example

A company can run an A/B test to determine whether a redesigned checkout page causes an increase in completed purchases.

## Caution

Causal claims require careful study design and assumptions; simple correlations are insufficient.

# 22. Diagnostic and Root-Cause Techniques

## Purpose

Root-cause analysis focuses on identifying underlying factors responsible for a problem rather than merely describing its symptoms.

## Techniques

The 5 Whys, fishbone diagrams, Pareto analysis, process mapping, segmentation, and variance decomposition can be useful.

## Example

If website conversion falls, analysts can investigate traffic source, device type, page speed, checkout errors, pricing, and campaign changes.

## Outcome

A good root-cause analysis should produce testable explanations and practical corrective actions.

# 23. Text and Sentiment Analysis

## Definition

Text analysis converts unstructured language into structured information that can be searched, categorized, summarized, or modeled.

## Methods

Keyword analysis, topic modeling, text classification, named-entity recognition, embeddings, and sentiment analysis are common.

## Applications

Customer reviews, support tickets, survey comments, news, and social-media text can be analyzed.

## Example

A service team can classify support messages by issue type and identify whether customer sentiment is positive, neutral, or negative.

# 24. Spatial and Geospatial Analysis

## Definition

Spatial analysis examines data associated with geographic locations, distances, shapes, and spatial relationships.

## Applications

Urban planning, disaster management, logistics, agriculture, public health, retail location planning, and environmental monitoring use spatial analysis.

## Techniques

Buffering, proximity analysis, spatial joins, hotspot analysis, interpolation, and geographic visualization are common.

## Example

A delivery company can identify areas with high order density and optimize delivery zones.

# 25. Real-Time and Streaming Analysis

## Definition

Real-time analysis processes data as it arrives or with very low delay. Streaming systems are useful when immediate decisions are important.

## Examples

Fraud alerts, sensor monitoring, network security, stock-market monitoring, and online recommendation systems can require streaming analysis.

## Challenges

High data velocity, latency, data quality, state management, scalability, and fault tolerance are important considerations.

## Outcome

The goal is often to detect an event or make a decision before delayed batch processing would be useful.

# 26. Data Analysis Tools

## Programming Tools

Python and R are widely used for statistical computing, visualization, machine learning, and automation. SQL is essential for querying structured databases.

## Spreadsheet Tools

Excel and similar spreadsheet applications are useful for smaller datasets, formulas, pivot tables, charts, and quick analysis.

## Visualization Tools

Power BI, Tableau, and other business-intelligence platforms help users explore and communicate analytical results.

## Libraries

Python users commonly work with pandas, NumPy, Matplotlib, Seaborn, scikit-learn, and specialized forecasting or deep-learning libraries.

# 27. Data Cleaning and Preparation

## Why It Matters

Poor-quality input data can produce unreliable analytical results. Cleaning is therefore an essential part of nearly every analysis project.

## Common Tasks

Handling missing values, removing duplicates, correcting inconsistent formats, treating outliers, validating ranges, encoding categories, and standardizing units are common.

## Example

A dataset containing dates in several formats should be standardized before time-based analysis.

## Best Practice

Document every important transformation so the analysis is reproducible and auditable.

# 28. Choosing the Right Analysis Method

## Decision Process

Start with the question, identify the outcome and predictors, determine variable types, inspect the data, consider assumptions, and select a method appropriate to the objective.

| Question | Possible Method |
|---|---|
| What happened? | Descriptive analysis |
| Why did it happen? | Diagnostic/root-cause analysis |
| What patterns exist? | EDA / clustering |
| Is there a relationship? | Correlation / regression |
| Is a difference statistically meaningful? | Hypothesis testing |
| What may happen next? | Predictive / time-series analysis |
| What action is best? | Prescriptive / optimization |

## Key Principle

The most sophisticated method is not always the best method. A simple, interpretable method may be preferable when it answers the question reliably.

# 29. Communicating Data Analysis Results & Summary, Best Practices and Review

## Purpose

Analysis has value only when its findings can be understood and used by decision-makers.

## Good Communication

Start with the main finding, provide supporting evidence, show appropriate visualizations, explain uncertainty, and clearly state implications.

## Visualization Principles

Choose charts according to the data and question. Avoid misleading scales, excessive decoration, unnecessary 3D effects, and clutter.

## Example

Instead of presenting a large table of monthly sales, a line chart can show the trend while a short summary highlights the key increase or decrease.

## Summary

Data analysis includes multiple complementary approaches. Descriptive analysis explains what happened; diagnostic analysis investigates why; exploratory analysis discovers patterns; inferential analysis generalizes from samples; predictive analysis estimates likely outcomes; and prescriptive analysis supports decisions.

## Best Practices

Define the question clearly, understand the data, clean and validate it, visualize before modeling, select methods based on the problem, check assumptions, quantify uncertainty, avoid unsupported causal claims, document the workflow, and communicate results honestly.

## Final Review Questions

1. What is the difference between descriptive and diagnostic analysis? 2. When is predictive analysis useful? 3. Why does correlation not prove causation? 4. What is the purpose of clustering? 5. How does qualitative analysis differ from quantitative analysis? 6. Why is data cleaning important? 7. Give two applications of time-series analysis.

## Learning Outcome

After completing this module, a learner should be able to identify major types of data analysis, distinguish their purposes, select suitable analytical approaches for common problems, and communicate analytical findings responsibly.